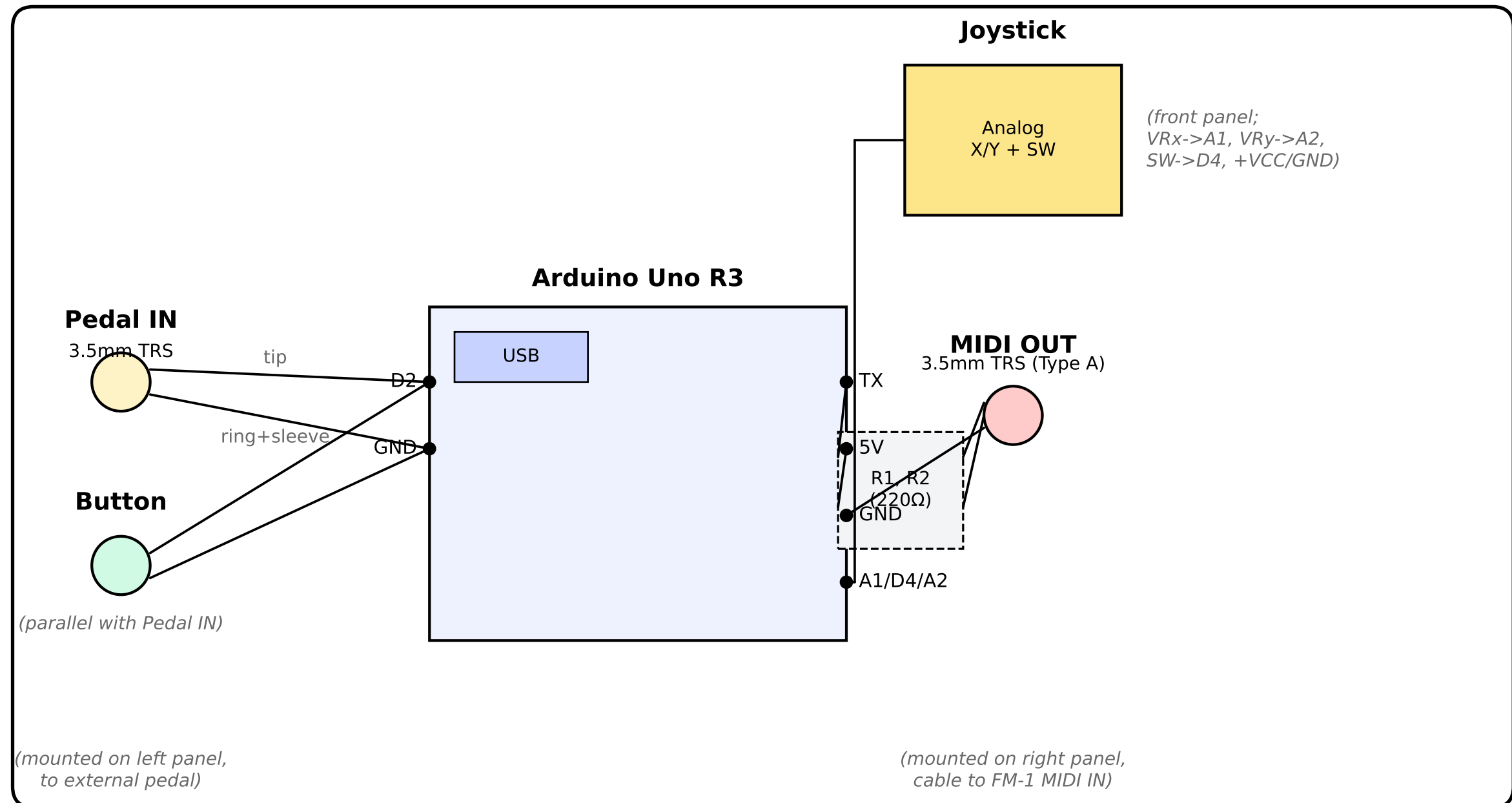


FM-1 Sustain Footswitch — Layout / Assembly Diagram

Top-down view of enclosure: pedal jack + built-in button (left panel), Arduino Uno (center), MIDI out jack (right panel), joystick (front panel)

project box (top cover removed)



Notes:

- Resistors R1/R2 can live on a small offcut of perfboard, or be soldered directly in-line on the wire runs -- exact placement isn't critical
- See the schematic for the electrical connections (which wire goes through which resistor) -- this diagram is for physical placement only
- Button and Pedal IN jack are wired in parallel to the same D2/GND nodes -- either one alone triggers sustain, so the pedal can be left unplugged
- Leave slack on the USB cable path -- the board still needs to be reachable for re-flashing (disconnect MIDI OUT jack's TX lead first)
- Joystick needs 5 wires to the Uno (VCC, GND, VRx->A1, VRy->A2, SW->D4) in addition to the single bundled point shown here for clarity
- OLED display, mode switch, and audio-input pitch detector removed for now -- coming back later